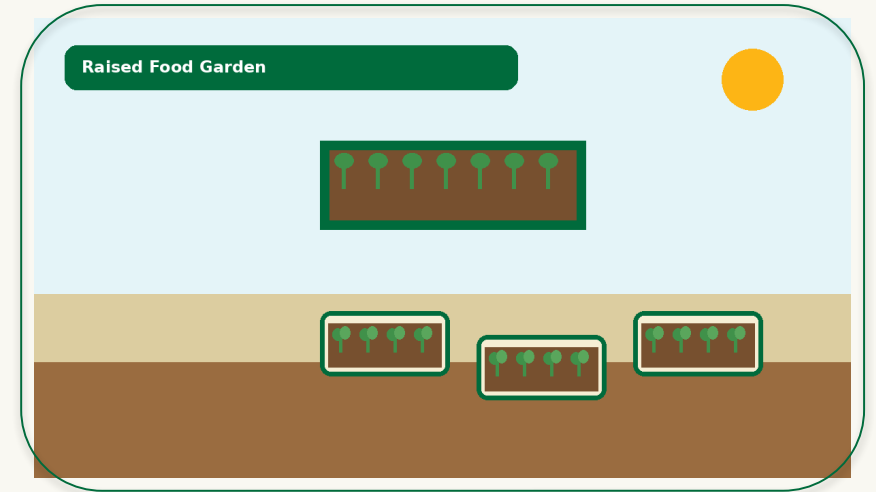


Raised Food Garden

Managing Productive Raised Beds

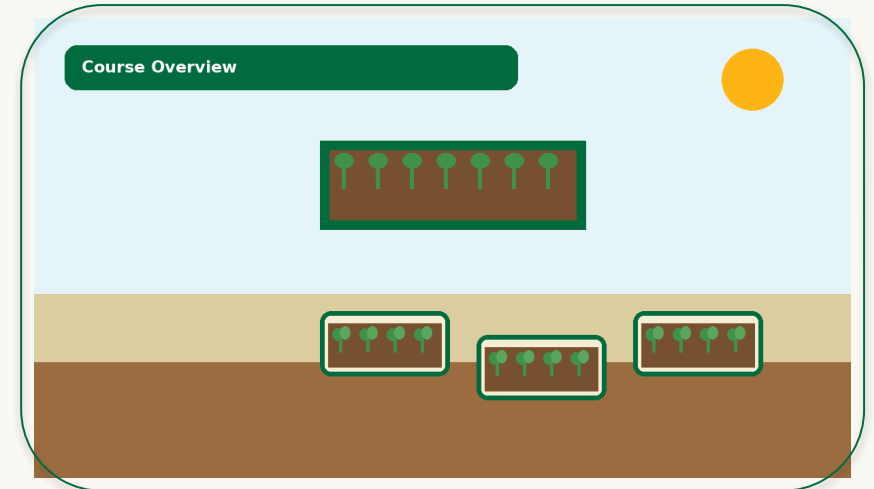
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Course emphasis: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden system and its educational purpose; teach the ten technical topics at Primary 4-6 level; connect mathematics/science/agriculture to practical work; supervise measurements and garden tasks safely; maintain usable records; guide pupils from observation toward evidence-based conclusions.
- Core materials: teacher guide, crop plan, child-safe tools, watering equipment, labels, clipboards, rulers/tapes, scales or thermometers where relevant, observation/data sheets, clean containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson/measurement or investigation plan.



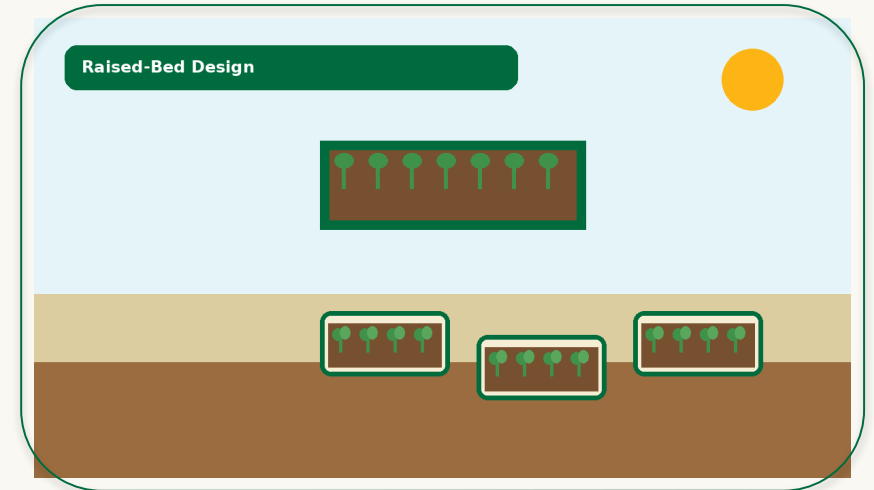
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Raised-Bed Design

Teacher knowledge and practical application

- Identify the functional parts of a raised bed: growing area, structural edge, path, soil profile and irrigation access.
- Discuss bed width in relation to safe reach so pupils can work without stepping into the soil.
- Explain how bed height, drainage and materials should suit the school site and crop plan.
- Compare rectangular, keyhole-like and modular layouts without claiming one design is universally best.
- Inspect structural stability, sharp edges and path access before student use.



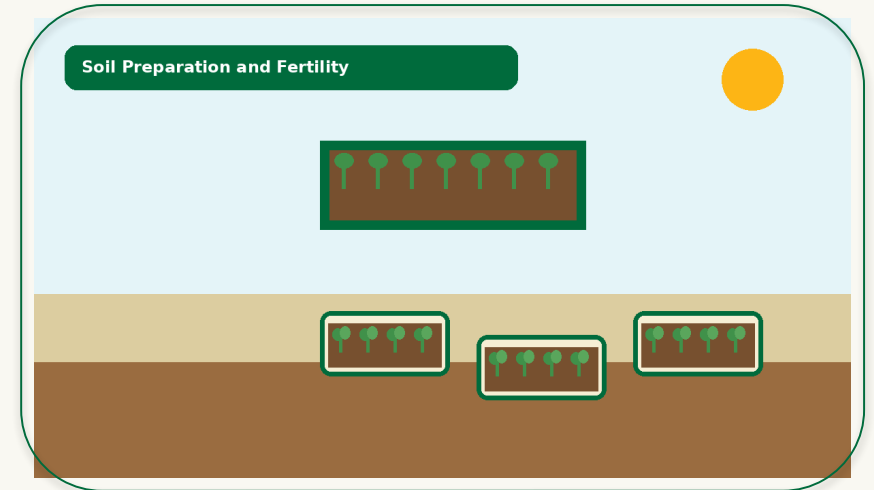
TEACHER PRACTICE

- Teacher task: evaluate one school raised bed for access, stability, drainage and teaching use.

Soil Preparation and Fertility

Teacher knowledge and practical application

- Explain soil as the root environment that supplies water, air and nutrients.
- Use mature compost and approved amendments according to the garden management plan.
- Demonstrate loosening, leveling and incorporation without unnecessary disturbance.
- Discuss simple fertility indicators while avoiding diagnosis from appearance alone.
- Keep fresh manure, concentrated fertilizers and unknown materials away from pupil handling.



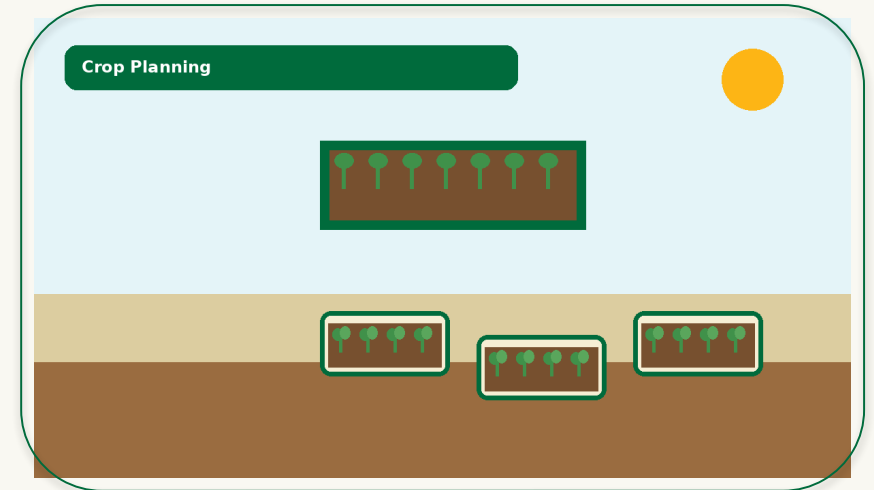
TEACHER PRACTICE

- Prepare a safe soil-preparation demonstration and fertility checklist.

Crop Planning

Teacher knowledge and practical application

- Select crops using season, available space, time to harvest, school calendar and learning objectives.
- Group crops by growth habit and approximate duration.
- Plan succession where feasible so beds remain productive across the term.
- Avoid planting more than teachers and pupils can maintain.
- Use a simple bed map showing crop, planting date and expected harvest window.



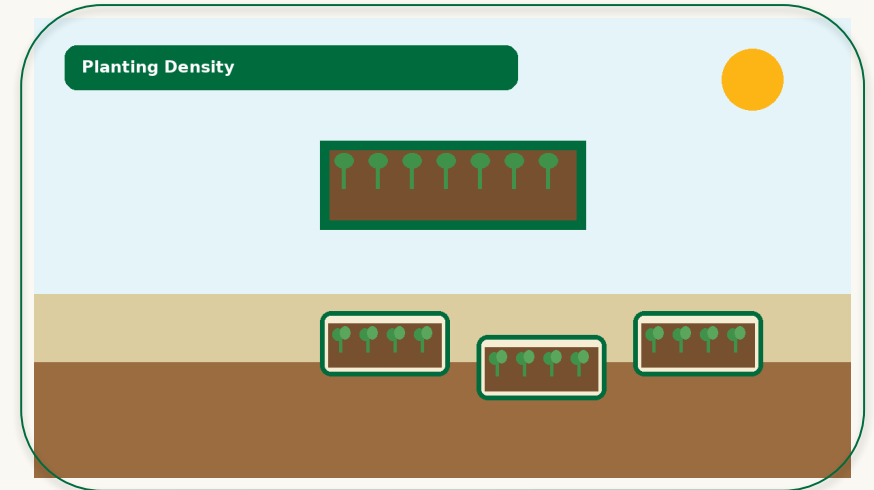
TEACHER PRACTICE

- Create a one-bed crop plan for a school term.

Planting Density

Teacher knowledge and practical application

- Define planting density as the number of plants occupying a given growing area.
- Use crop-specific spacing recommendations rather than arbitrary patterns.
- Show how excessive crowding can affect access, airflow and plant development.
- Calculate simple plant counts from rows and spacing at an age-appropriate level.
- Recheck established stands and record missing or overcrowded positions.



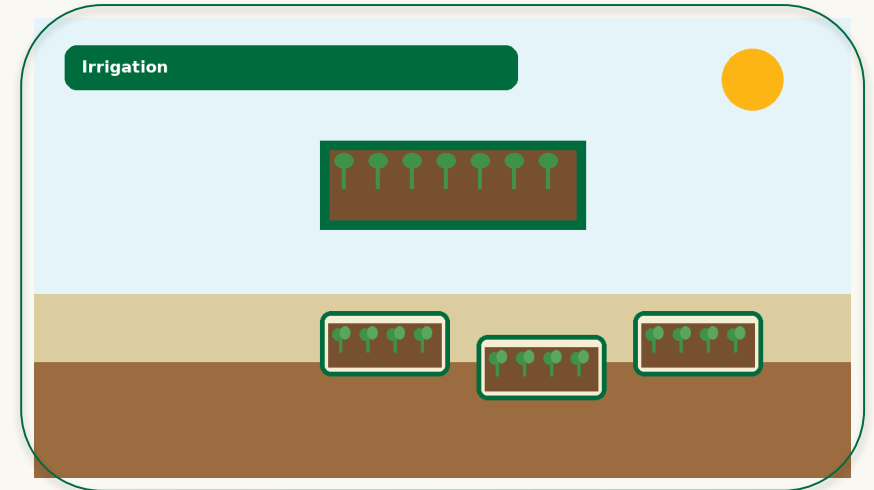
TEACHER PRACTICE

- Prepare a spacing-and-density activity using one crop.

Irrigation

Teacher knowledge and practical application

- Identify the water source, delivery method and root zone.
- Compare hand watering and drip irrigation where both are available.
- Check soil moisture before watering and explain why fixed schedules may need adjustment.
- Apply water without creating runoff, erosion or unsafe wet paths.
- Record major irrigation events for later comparison with plant performance.



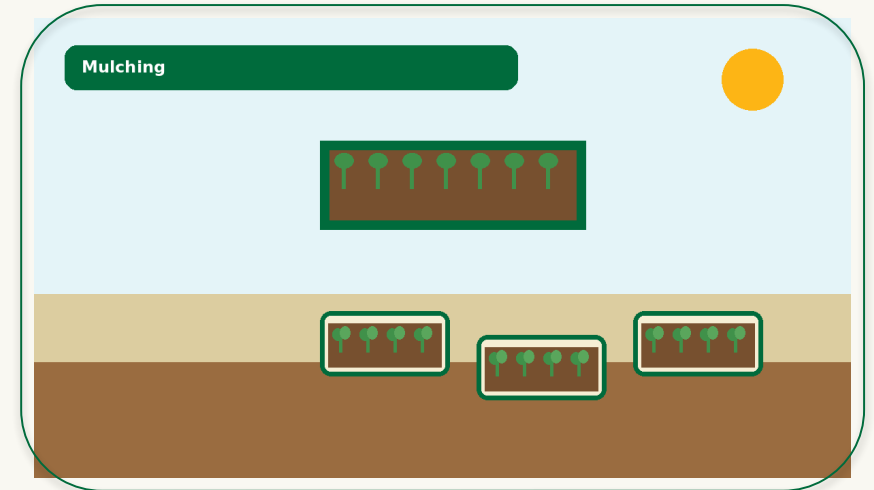
TEACHER PRACTICE

- Demonstrate CHECK -> APPLY -> OBSERVE -> RECORD.

Mulching

Teacher knowledge and practical application

- Explain mulch as material placed on the soil surface for functions such as reducing evaporation, moderating surface conditions and suppressing some weeds.
- Use clean, approved mulch materials.
- Keep mulch away from direct contact with vulnerable stems where appropriate.
- Compare a mulched and unmulched surface through observation rather than promising a fixed result.
- Monitor for pests, excessive moisture or decomposition issues.



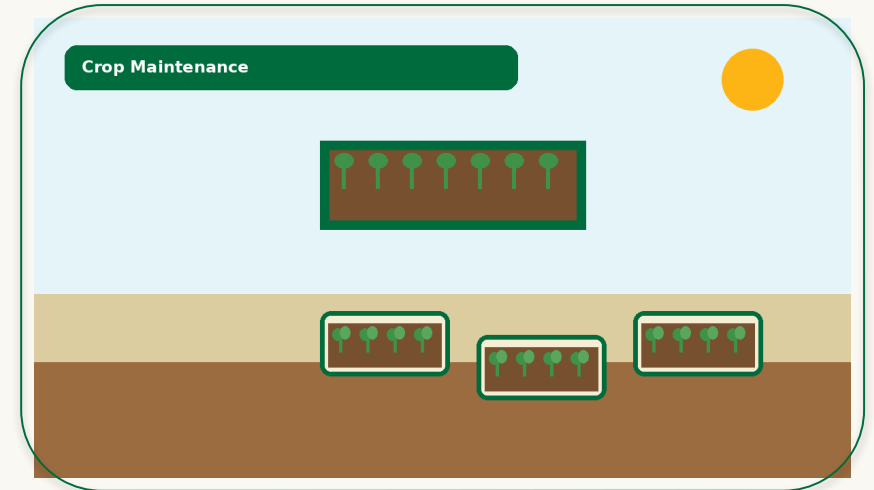
TEACHER PRACTICE

- Set up a simple mulch observation comparison.

Crop Maintenance

Teacher knowledge and practical application

- Organize weeding, thinning, staking, pruning where crop-appropriate, irrigation checks and removal of damaged material.
- Teach pupils to identify the crop before removing nearby plants.
- Use age-appropriate tools and clear work zones.
- Maintain labels, paths and irrigation components.
- Record significant maintenance actions.



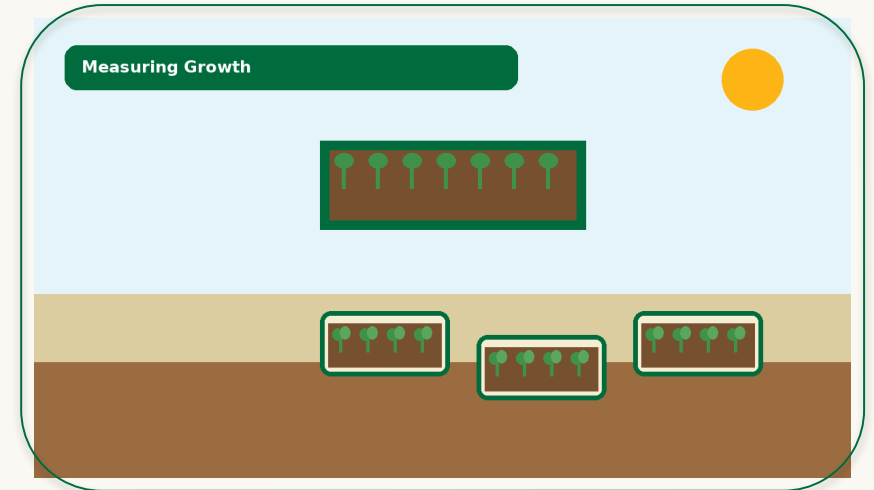
TEACHER PRACTICE

- Develop a weekly raised-bed maintenance rota.

Measuring Growth

Teacher knowledge and practical application

- Choose consistent indicators such as plant height, leaf number or canopy width.
- Define exactly how and where each measurement is taken.
- Use rulers or tapes correctly and record units.
- Measure the same marked plants across dates when tracking change.
- Avoid damaging plants merely to obtain measurements.



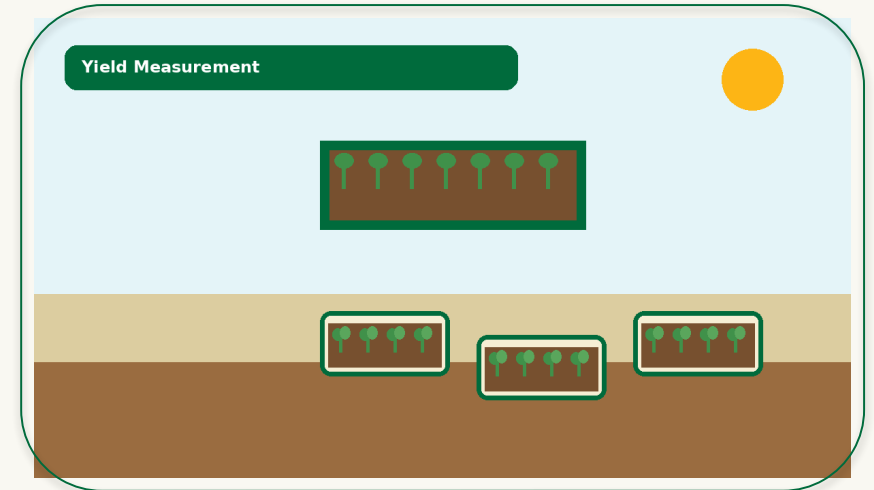
TEACHER PRACTICE

- Prepare a growth-monitoring sheet for five marked plants.

Yield Measurement

Teacher knowledge and practical application

- Define yield for the lesson as harvested crop quantity from a known bed or area.
- Count items and/or measure mass with a suitable scale.
- Record the bed, crop, date and unit.
- Use consistent harvest criteria when comparing beds.
- Explain that one harvest measurement does not by itself explain why yields differ.



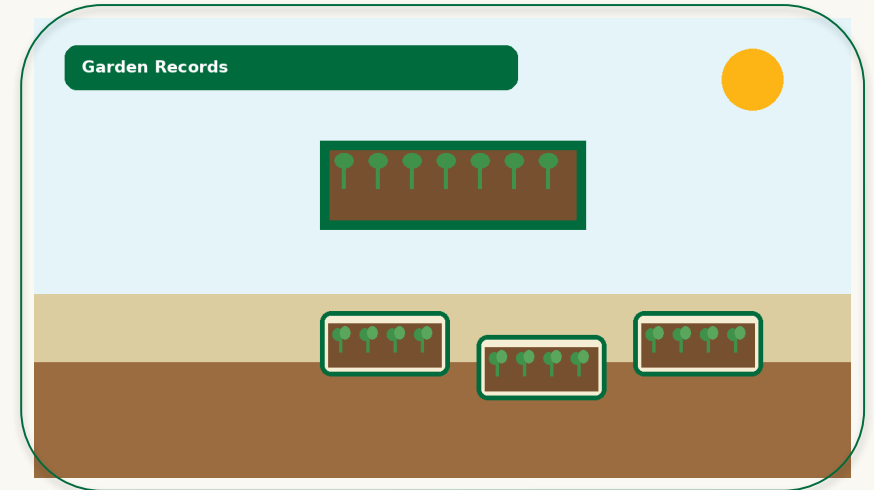
TEACHER PRACTICE

- Prepare a harvest count-and-weigh procedure.

Garden Records

Teacher knowledge and practical application

- Maintain a simple record of crop, planting date, bed, inputs/care, growth observations and harvest.
- Use consistent dates and units.
- Separate direct observations from explanations or guesses.
- Use records to discuss what changed and what management action may be needed.
- Store records so future classes can compare seasons.



TEACHER PRACTICE

- Completion task: create a complete raised-bed record and 40-minute teaching lesson.